Abhijit Kumar

Software Engineer @ Flipkart

CONTACT

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SKILLS

CORE

DSA · System Design · Microservices · Distributed Systems · SDLC · REST API Design · CI/CD

LANGUAGES

Java · Python · C++ · SQL

AI / ML

Agenitic skill / MCP · Prompt engineering · Data Science · Deep Learning · Reinforcement Learning · RAG

FRAMEWORKS & DB

Spring Boot · REST APIs · Dynamic Quartz · MySQL · TiDB · MongoDB · Aerospike · Redis

INFRASTRUCTURE

Docker · Kubernetes · Kafka · Prometheus · Grafana

TOOLS

Git · Jenkins · Linux · Cursor · MCP

SUMMARY

Software Engineer at Flipkart with experience building and scaling distributed microservices. Have taken features end-to-end — from solutioning and HLD/LLD through cross-team contract discussions, NFR testing, and production rollout. Comfortable working with high-QPS systems, resolving latency and concurrency issues, and handling large-scale data operations. Actively using AI-assisted tooling in day-to-day workflows and learning the tech behind it.

EDUCATION

National Institute of Technology, Durgapur — 2020 – 2024

B.Tech, Computer Science and Engineering — 8.84 CGPA

EXPERIENCE

SDE 2, Flipkart — Feb 2024 – Present

SDE Intern: Feb 2024 – May 2024, SDE1: May 2024 – Jan 2026

- **Scaled Flipkart scratch card flow from `120 QPS` to `500 QPS` for Big Billion Days** — this flow had to be disabled in the previous sale due to throughput limits.
- **SuperCoin Earn-as-an-Offer e2e (Coins earn visible on flipkart product page):** Solutioning, HLD/LLD, API contract with multiple teams & API integration, SuperCoin library changes, analytics support, integration testing, and deployment across the order funnel.
- **Built an MCP server** for MySQL and TiDB database interaction, allowing the team to query and debug databases directly through Cursor. Adopted across the entire Lockin team and used daily for oncall debugging and rca.
- **Delivered the CnC NEP Revamp (Visible as Wow deal on Flipkart product page)** — revamped the coin distribution system by along with contract changes with 3 external teams, rebuilding the coins calculation library and service, and running Load tests.
- **Handled large-scale data operations:** ingested `1.54B records (~1 TB)` into Khulasa reporting for Beagle onboarding; wrote fault-tolerant Python multiprocessing scripts with retry mechanisms to reconcile `31M` Cleartrip accounts at `1.2k QPS`.
- **Built a dynamic rate limiting system** using Kubernetes ConfigMap listeners in a shared utility library, removing the need for service restarts when onboarding new clients to high-traffic APIs.
- **Designed Database query based monitoring service with scheduling support,** a policy monitoring service with dynamic Quartz scheduling and SQL-based policy execution across database clusters. Set up the full Kubernetes pre-prod environment from scratch.
- **Handled critical production incidents** including a SEV-0 SuperCoins outage (30% 5XX rate across two DCs), Reward service OOM failures, and service availability drops. Wrote RCAs and shipped long-term fixes. Created multiple SOPs.
- **Mentored 2 SDE1 (new hire) and 1 SDE intern**

Research Intern, TCS Research Lab — May – Jul 2023

- Built real-time object detection and segmentation models using deep learning. Created annotated datasets for the CTO team, enabling autonomous robot segmentation in real-time scenarios.

ACHIEVEMENTS

- **3 Instant Karma and 1 Ace Aliance award at Flipkart**
- **National Finalist, Flipkart Grid 5.0** — Top 0.01% of 4.7 Lakh+ teams
- **1st Place** — App Development Hackathon, IIT Madras (15,000+ teams)
- **1st Place** — Research Hackathon for Telescope Observation Simulation, DTU

PROJECTS

Q-Learning for Agentic Optimization: RL in Multi-Objective Pathfinding

Formulated navigation as MDPs with value-based Q-Learning to build a Multi-Source Shortest Path model outperforming deterministic methods. Built an autonomous agent (Tour de Flag) using epsilon-greedy exploration and TD updates for obstacle avoidance and multi-objective flag collection.

YOLO Instance Segmentation: "Simple Copy-Paste" Data Augmentation

Implemented a data augmentation pipeline for YOLO-formatted instance segmentation datasets based on the CVPR 2021 "Simple Copy-Paste" paper to improve model generalization.

Hybrid Classical-Quantum Neural Network for Diabetic Retinopathy

Quantum-enhanced deep CNN improving classification accuracy of diabetic retinopathy progression on IDRID Dataset, outperforming conventional baselines.

SnapMap — Geotagged Image Capture with Heatmap Visualization

Mobile app for real-time geotagged image capture, cloud storage, and heatmap visualization of user activity.